

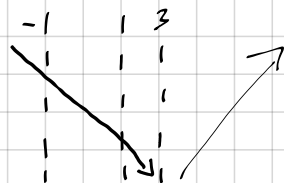
D1

$$y = x^3 - 3x \quad I = (-1, 1)$$

$$y' = 3x - 3$$

$$3x - 3 \geq 0 \Rightarrow x - 3 \geq 0 \Rightarrow x \geq 3$$

in I , y is decreasing



DZ

$$y = \ln(4 - x^2)$$

$$4 - x^2 > 0 \Rightarrow -x^2 > -4 \Rightarrow x^2 < 4 \Rightarrow -2 < x < 2$$

$$D = (-2, 2)$$

D 4

$$\int_1^2 \frac{1}{x} + 2x = \left[\ln(x) + x^2 \right]_1^2 = (\ln(2) + 4) - (\ln(1) + 1) = (\ln(2) + 4) - 1 = \ln(2) + 3$$

$$\int \frac{1}{x} + 2x \, dx = \int \frac{1}{x} \, dx + \int 2x \, dx = \ln(x) + 2 \int x \, dx = \ln(x) + 2 \frac{x^2}{2} = \ln(x) + x^2 + C$$

D 7

$$y = x^3 + x + 1$$

D8

$$f = \begin{cases} \sin(x^2) + 0 & x \leq 0 \\ \frac{\ln(1+x)}{x+x^3} & x > 0 \end{cases}$$

$$\lim_{x \rightarrow 0^+} \frac{\ln(1+x)}{x+x^3} \stackrel{H}{=} \frac{1}{x+1} \cdot 1+3x^2 = 1 \cdot 1 = 1$$

$$\lim_{x \rightarrow 0^-} \sin(x^2) + 0 = 0$$

$$0 = 1$$

D.5

$$\lim_{n \rightarrow +\infty} (n^2 - n^3 + 3e^{-n} + \cos n^2)^2 - n^3 = -\infty$$

D 6

$$A = \left\{ \frac{z}{z^2+1} + e^{1-n} - 1 \right\}$$

$$\lim_{n \rightarrow +\infty} \frac{z}{z^2+1} + e^{1-n} - 1 = -1$$

$$\frac{z}{z \cdot 1 + 1} + e^0 - 1 = \frac{z}{3}$$

D3

$$\sum 4^{-n}$$

$$\frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{4} : \frac{4-1}{4} = \frac{1}{4} : \frac{3}{4} = \frac{1}{4} \cdot \frac{4}{3} = \frac{1}{3}$$

Eg 1

$$y = (x^2 - x) e^{-x}$$

DOMINIO

$$D = \mathbb{R}$$

LIMITI

$$\lim_{x \rightarrow +\infty} (x^2 - x) e^{-x} = \frac{(x^2 - x)}{e^x} = 0 \quad \text{opp. orizzontale}$$

$$\lim_{x \rightarrow -\infty} (x^2 - x) e^{-x} = \frac{(x^2 - x)}{e^x} = +\infty \quad \text{opp. obliqua?}$$

$$\lim_{x \rightarrow -\infty} \frac{(x^2 - x) e^{-x}}{x} = \frac{(x^2 - x)}{e^{-x} x} = \frac{x(x-1)}{x} \cdot \frac{1}{e^x} = \frac{x-1}{e^x} = +\infty \quad \text{no opp. obliqua}$$

SEGNO

$$y = (x^2 - x) e^{-x} = \frac{x^2 - x}{e^x}$$

$$\frac{x^2 - x}{e^x} > 0 \quad \left\{ \begin{array}{l} x^2 - x > 0 \\ e^x > 0 \end{array} \right\} \quad \left\{ \begin{array}{l} x(x-1) > 0 \\ \forall x \end{array} \right\} \quad \left\{ \begin{array}{l} x > 0 \\ x-1 > 0 \end{array} \right\} \quad \left\{ \begin{array}{l} x > 0 \\ x > 1 \end{array} \right\} \Rightarrow x > 1$$

y positive per $x \in (1, +\infty)$, y negative per $x \in (-\infty, 1)$

MONOTONIA

$$y = (x^2 - x) e^{-x}$$

$$y' = (2x - 1)(e^{-x}) + (x^2 - x)(-e^{-x}) = 2x e^{-x} - e^{-x} - x^2 e^{-x} + x e^{-x} = e^{-x}(-x^2 + 3x - 1)$$

$$e^{-x}(-x^2 + 3x - 1) > 0 \Rightarrow -x^2 + 3x - 1 > 0 \Rightarrow \frac{3 - \sqrt{5}}{2} < x < \frac{3 + \sqrt{5}}{2}$$

MAX & MIN

$$e^{-x}(-x^2 + 3x - 1) = 0$$

$$-x^2 + 3x - 1 = 0$$

$$x^2 - 3x + 1 = 0$$

$$x_1 = \frac{3 - \sqrt{5}}{2} \quad x_2 = \frac{3 + \sqrt{5}}{2}$$

x_1 punto di minimo, x_2 punto di massimo

CONCAVITA' & CONVISSITA'

$$y = (x^2 - x)e^{-x}$$

$$y' = e^{-x}(-x^2 + 3x - 1)$$

$$y'' = (-e^{-x})(-x^2 + 3x - 1) + (e^{-x})(-2x + 3)$$

$$x^2 e^{-x} - 3x e^{-x} + e^{-x} + (-2x e^{-x} + 3e^{-x})$$

$$x^2 e^{-x} - 3x e^{-x} + e^{-x} - 2x e^{-x} + 3e^{-x}$$

$$x^2 e^{-x} - 5x e^{-x} + 4e^{-x}$$

$$e^{-x}(x^2 - 5x + 4)$$

$$e^{-x}(x^2 - 5x + 4) > 0 \Rightarrow x^2 - 5x + 4 > 0 \Rightarrow (-\infty, 1) \cup (4, +\infty)$$

5)

$$\int_0^1 \frac{(x^2 - x)e^{-x}}{x} = \left[e^{-x}(-x + 2) \right]_0^1 = (e^{-1}(-1 + 2)) - (e^0(0 + 2)) = (e^{-1}(1)) - (2) =$$

$$\int_0^1 \frac{(x^2 - x)e^{-x}}{x} = \frac{x^2 - x}{x} \cdot e^{-x} = \frac{x(x - 1)}{x} \cdot e^{-x} = (x - 1)e^{-x} = x e^{-x} - e^{-x} = \int x e^{-x} dx - \int e^{-x} = \int x e^{-x} - \int e^y =$$

$$\begin{aligned} y &= -x \\ dy &= -dx \\ -dy &= dx \end{aligned}$$

$$\int x e^{-x} - \int e^y = \int x e^{-x} - e^y = \int x e^{-x} - e^{-x} = -x e^{-x} - \int -e^{-x} - e^{-x} = -x e^{-x} - e^{-x} - e^{-x} + C = -x e^{-x} - 2e^{-x} + C$$

$$-x e^{-x} - 2e^{-x} + C = e^{-x}(-x + 2) + C$$